Treasury Lock Model:
A Practitioner's Forward-Bond Approach

Anonymous (n.d.) – Working Notes

Setting and Goal

The paper is a short, practical T-Lock pricing manual: four pages describing how a desk values a long T-Lock as a forward contract on the underlying treasury, with three lock conventions (clean price, yield, dirty price). It complements the Pucci T-Lock paper (SSRN 3386521) already in our reference set, which derives the trade's payoff, convexity, and overhedge structure from first principles.

The two papers are best read together:

- **Pucci (2019)** – *theoretical framework*: the T-Lock as a forward IRR contract, exact payoff $N \cdot (\mathrm{IRR} - L) \cdot \mathrm{RiskFactor}_{t_e}$, the forward-contract overhedge approximation, negative convexity, hedge-cost analysis.
- **This paper** – *operational pricing recipe*: discrete coupon-by-coupon forward formula on the repo curve, three lock conventions, sensitivities. The recipe a desk would actually use day-to-day.

The goal of these notes is to (i) restate the paper's three pricing formulas precisely, (ii) make the bond-forward derivation auditable from a cash-and-carry argument, (iii) reconcile the practitioner's PV01-based yield-lock formula with Pucci's exact forward-IRR expression, and (iv) generate a validation suite for our pricing library.

1 Trade

A T-Lock is a customised forward contract that fixes the yield or price on a specified treasury security between trade date t and delivery T_{del}. Lock terms are typically one week to twelve months. The buyer is protected against a yield rise (price fall) over the lock period; the seller takes the opposite exposure.

Per the paper's introduction, the model inputs are:

- the coupon and clean / dirty price of the underlying bond;
- the market quote (used as P_0 when `BOND_FROM_QUOTE = true`);
- the accrued interest at t and at T_{del};
- the repo curve, used to discount coupons paid before delivery;
- the bond zero-yield curve, used to discount the T-Lock's terminal payoff back to t.

The bond zero-yield curve referenced for the T-Lock discount is the "FiZeroBond" construction described separately by the same author (see `finpricing.com/lib/FiZeroBond.html`, cited inline in the paper).

2 Bond Forward Price – Cash-and-Carry on the Repo Curve

2.1 Statement

Proposition 1 (Bond forward, clean price)**.** *Let P_0 be the spot clean price, A_0 the accrued interest at spot settlement T_0, and A_{del} the accrued interest at delivery T_{del}. Let C_i be the coupons paid at intermediate dates T_i, $i = 1, \ldots, N$, with $T_0 < T_1 < \cdots < T_N \leq T_{\text{del}}$. Let $D_{\text{repo}}(s, T)$ be the repo discount factor from T back to s. Then the bond forward price (clean) at delivery is*

$$\boxed{B_f \;=\; \frac{P_0 + A_0}{D_{\text{repo}}(T_0, T_{\text{del}})} \;-\; \sum_{i=1}^{N} \frac{C_i}{D_{\text{repo}}(T_i, T_{\text{del}})} \;-\; A_{\text{del}}.} \tag{1}$$

2.2 Cash-and-carry derivation

The formula is the no-arbitrage cost of replicating bond delivery at T_{del} via repo financing:

1. At T_0, borrow cash on repo collateralised by the bond: the financing cost grown to T_{del} is $(P_0 + A_0)/D_{\text{repo}}(T_0, T_{\text{del}})$, in dirty-price terms.
2. Each intermediate coupon C_i paid at T_i is reinvested at the repo rate to T_{del}: total coupon proceeds at delivery are $\sum_i C_i/D_{\text{repo}}(T_i, T_{\text{del}})$, which reduces the financing balance.
3. At delivery the position is closed by paying the dirty forward price; converting to a clean quote subtracts A_{del}.

Remark 1 (Connection to Pucci's continuous-yield formula)**.** *Pucci (SSRN 3386521, Appendix D) writes the forward in the continuous-coupon-yield form $B_f^{\text{Pucci}} = P_0\, e^{(r_{\text{repo}} - c)(T_{\text{del}} - T_0)}$, where c is the continuously-compounded coupon yield. Equation (1) is the discrete-coupon analogue: each C_i is treated as a discrete cash flow rather than a continuous yield. The two formulas agree to first order when $C_i = c\, P_0\, \Delta_i$ and $D_{\text{repo}}(T_i, T_{\text{del}}) \approx 1 - r_{\text{repo}}(T_{\text{del}} - T_i)$.*

Remark 2 (Default selection of P_0)**.** *The paper notes that with `BOND_FROM_QUOTE = true` (the default), P_0 is set to the available market quote. Otherwise the clean price is computed from the bond's zero-yield curve. This makes B_f market-consistent when the underlying is liquid, and model-consistent (yield-curve based) when it is not – a small but non-trivial implementation choice that the validation suite must respect.*

2.3 Forward yield y_f

The corresponding forward IRR y_f is the unique solution of

$$B_f + A_{\text{del}} \;=\; \sum_j \frac{c_j}{(1 + y_f/k)^{k\,\tau_j}}, \tag{2}$$

where the sum is over all bond cash flows c_j paid after T_{del}, with τ_j the year-fraction from T_{del} to the cash-flow date and k the compounding frequency. Equation (2) is the standard bond-pricing equation applied to the forward dirty price; y_f is therefore a function of the inputs (P_0, A_0, A_{del}, repo curve, coupons) via B_f.

3 The Three Lock Conventions

3.1 Lock the clean price

Proposition 2 (Clean-price T-Lock)**.** *Let $P_{\text{T-Lock}}$ be the locked clean price, N the bond quantity, and $D_{\text{T-Lock}}(t, T_{\text{del}})$ the discount factor from T_{del} back to valuation date t on the bond zero-yield curve. The buyer's NPV is*

$$\boxed{\text{NPV}_{\text{T-Lock}} \;=\; N\left(P_{\text{T-Lock}} - B_f\right) D_{\text{T-Lock}}(t, T_{\text{del}}).} \tag{3}$$

Remark 3 (Sign convention)**.** *The buyer's NPV is positive when the locked price exceeds the forward, i.e. the buyer profits if the forward (and hence the expected delivery price) drops below* $P_{\text{T-Lock}}$*. This is consistent with the buyer being protected against a price fall.*

3.2 Lock the yield

Proposition 3 (Yield T-Lock)**.** *Let* $y_{\text{T-Lock}}$ *be the locked yield,* M *the trade amount,* y_f *the forward yield from* (2)*, and* $B(y)$ *the present value of the unit underlying bond at yield* y*. Define*

$$\text{PV01}(y_f) \;\coloneqq\; B\big(y_f + \tfrac{1}{2}\text{bp}\big) \;-\; B\big(y_f - \tfrac{1}{2}\text{bp}\big)\,. \tag{4}$$

The buyer's NPV is

$$\boxed{\text{NPV}_{\text{T-Lock}} \;=\; M\,(y_f - y_{\text{T-Lock}})\,\big|\text{PV01}(y_f)\big|\,D_{\text{T-Lock}}(t, T_{\text{del}}).} \tag{5}$$

Remark 4 (Sign convention – yield case)**.** *The buyer's NPV in* (5) *is positive when* $y_f > y_{\text{T-Lock}}$*, i.e. when forward yields have risen above the locked yield. This is the price-fall / yield-rise scenario the buyer is hedging against, consistent with* (3)*.*

Remark 5 (PV01(y_f) vs. standard PV01)**.** *The paper states explicitly that* PV01(y_f) differs *from the standard* PV01*. Two distinctions:*

1. *It is evaluated at the* forward *yield* y_f*, not the spot IRR.*
2. *It is a centred finite difference on* $B(\cdot)$ *(Equation* (4)*) – a numerical, not analytical, risk factor.*

The risk factor in (5) *therefore corresponds to Pucci's* RiskFactor_{t_e} *at* $t_e = T_{\text{del}}$*, evaluated at the forward IRR, with absolute value taken so that the sign of the NPV is carried entirely by* $(y_f - y_{\text{T-Lock}})$*.*

3.3 Lock the dirty price

For a dirty-price lock $P^{\text{dirty}}_{\text{T-Lock}}$, set $P_{\text{T-Lock}} = P^{\text{dirty}}_{\text{T-Lock}} - A_{\text{del}}$ and apply the clean-price formula (3). The paper treats this as a notational rather than structural variant.

3.4 Discounting choice – repo curve vs. bond yield curve

Remark 6 (Why two curves)**.** *Two curves play distinct roles, and the paper is precise about which appears where:*

- *the* repo curve *prices the cash-and-carry that determines* B_f *(Equation* (1)*) – it is the cost of financing the hedge between* T_0 *and* T_{del}*;*
- *the* bond zero-yield curve *discounts the T-Lock terminal payoff from* T_{del} *back to* t *(Equations* (3) *and* (5)*) – it is the rate at which the forward-fixed cash flow is valued.*

This is the same drift-versus-discount distinction that appears in the discounting framework paper (see Section 7): the repo rate enters drift *(forward price); a separate curve enters* discount*. Conflating them is a common bug.*

4 Reconciliation with the Pucci Framework

Pucci writes the long-T-Lock payoff at expiry $t_e = T_{\text{del}}$ as

$$\text{Payoff} \;=\; N \cdot (\text{IRR} - L) \cdot \text{RiskFactor}_{t_e}, \tag{6}$$

where RiskFactor_{t_e} is the IRR-sensitivity of the bond price computed at t_e. Pucci's main result is that the T-Lock *at first order* in $(\text{IRR} - L)$ is approximated by a short forward struck at $P_{t_e}(L)$, and that this approximation is an overhedge because the T-Lock has negative convexity (Pucci, eq. (43)).

Proposition 4 (Reconciliation)**.** *The paper's yield-lock formula* (5) *is exactly the t-PV of Pucci's payoff* (6) *under the two operational approximations:*

1. *the forward IRR is taken as deterministic, equal to its t-forward y_f (no convexity adjustment);*
2. RiskFactor_{t_e} *is replaced by its t-forward estimate* $|\text{PV01}(y_f)|$.

Specifically,

$$\mathbb{E}^{\mathbb{Q}}[N \cdot (\text{IRR} - L) \cdot \text{RiskFactor}_{t_e} \,|\, \mathcal{F}_t] \; D_{\text{T-Lock}}(t, T_{\text{del}}) \xrightarrow{(1),(2)} M\,(y_f - y_{\text{T-Lock}})\,\big|\text{PV01}(y_f)\big|\, D_{\text{T-Lock}}(t, T_{\text{del}}), \tag{7}$$

with $M = N$ and $y_{\text{T-Lock}} = L$.

Remark 7 (What is dropped)**.** *The first-order approximation drops the convexity correction arising from the joint randomness of* IRR *and* RiskFactor*, which Pucci shows is negative (the T-Lock is locally concave near the locked yield). The forward-contract overhedge identified in Pucci (page 8 onward of SSRN 3386521) is exactly the difference between the right-hand side of* (7) *and the true t-value of* (6)*.*

Remark 8 (When the approximation is acceptable)**.** *The paper's PV01-based formula is a desk-grade pricer, fast, and exact at trade inception when $y_{\text{T-Lock}} = y_f$ (zero NPV by design). It is most accurate over the short tenors $T_{\text{del}} - t \lesssim 6$ months that are typical of T-Lock pre-hedge transactions, where the unhedged convexity exposure is small. For longer tenors or out-of-the-money locks (large $|y_f - y_{\text{T-Lock}}|$), the convexity correction in the Pucci framework becomes material.*

5 Sensitivities – The Paper's Caveat

The paper explicitly notes that the T-Lock has *no specific risk indicator*: both PV01 and NDELTA available in the system are underlier-related, applied as approximations.

This honesty is significant. The genuine T-Lock risk drivers are:

- movement of the forward yield y_f (the locked variable itself);
- shape of the repo curve (which moves y_f via B_f);
- shape of the bond zero-yield curve (which moves the discount $D_{\text{T-Lock}}$);
- time decay through both the B_f formula and $D_{\text{T-Lock}}$.

The standard PV01 / NDELTA on the underlying bond captures only the first of these, and only with respect to spot rather than forward yield. A complete T-Lock risk view requires bumping the repo curve and the discount curve separately and observing the impact on B_f and $\text{NPV}_{\text{T-Lock}}$.

6 Worked Example – Yield Lock on a 10y Treasury

To make the magnitudes concrete, consider Pucci's 24 Jan 2019 example transposed to the present paper's formulas:

- Underlying: US Treasury, coupon 3.125%, maturity 25 Nov 2028 (ISIN US9128285M81);
- Spot clean price $P_0 = 103.4922$, dirty 104.1055;
- Spot IRR = 2.717%;

- 3m repo $r_{\text{repo}} = 2.46\%$, $T_{\text{del}} - T_0 = 0.25$y;
- Forward dirty $B_f + A_{\text{del}} = 104.74$, forward clean $B_f = 103.36$;
- Forward yield $y_f = 2.530\%$;
- Locked yield $y_{\text{T-Lock}} = 2.717\%$ (locked at spot).

For a unit notional ($M = 1$), the modified duration of a 10y bond at 2.5% yield is ≈ 8.5, so $|\text{PV01}(y_f)| \approx 0.085$ per unit. Therefore

$$\text{NPV}_{\text{T-Lock}} \;\approx\; (0.02530 - 0.02717) \times 0.085 \times D_{\text{T-Lock}}(t, T_{\text{del}}) \;\approx\; -1.59 \times 10^{-4}\, D_{\text{T-Lock}}(t, T_{\text{del}})$$

per unit notional. Scaled to a \$100m notional and assuming $D_{\text{T-Lock}} \approx 0.994$, the buyer's NPV is approximately $-$\$15,800, mostly from the carry-induced gap between y_f and the locked-at-spot yield. This is the order of magnitude consistent with Pucci's example.

7 Connection to the Other Papers in the Reference Set

Pucci T-Lock (SSRN 3386521). The framework paper for the trade analysed here. Section 5 above gives the explicit reconciliation of the present paper's PV01-based formula with Pucci's forward-IRR payoff. The forward bond formula (1) is the discrete-coupon analogue of Pucci's Appendix-D continuous-yield form.

Anonymous Discounting Paper (`What Textbooks Get Wrong...`). The role of the repo curve in the bond forward (1) is the *drift channel* of that paper's Section 3: the repo rate of the underlying bond determines the forward price, even while a different curve (here the bond zero curve) discounts the T-Lock cash flow. The two-curve structure of the T-Lock pricer is the simplest possible illustration of the drift-versus-discount distinction.

Pucci CMS / CMASW / CMT (SSRN 1961545, 2056277, 3387961). The CMT and T-Lock share the same underlying machinery (bond zero-yield curve, forward IRR, IRR-payoff structure). Pucci CMT extends to credit risk via the credit-risk-adjusted discount $\widehat{\widehat{D}}$, where here we use the unadjusted bond yield curve.

Lou TRS (SSRN 3217420). The TRS uses the repo rate of the underlying as the funding rate for the security leg, exactly as the T-Lock does in the bond-forward formula. In Lou's regime taxonomy, the T-Lock is a fully cash-settled forward; under a perfect CSA on the T-Lock itself, $D_{\text{T-Lock}}$ is the OIS curve.

8 Validation Notes for the Pricing Library

The validation suite below is designed so each test isolates a single component of the T-Lock pricer, and so that anchor numbers agree with Pucci where the two papers overlap.

8.1 Bond forward

1. **Zero-coupon bond limit.** Strip all coupons from the underlying. Equation (1) reduces to $B_f = P_0/D_{\text{repo}}(T_0, T_{\text{del}}) - A_{\text{del}}$. With $A_0 = A_{\text{del}} = 0$, this is the trivial cash-and-carry. Verify that the library returns exactly P_0/D_{repo}.
2. **Single intermediate coupon.** One coupon C_1 at T_1. Verify $B_f = (P_0 + A_0)/D_{\text{repo}}(T_0, T_{\text{del}}) - C_1/D_{\text{repo}}(T_1, T_{\text{del}}) - A_{\text{del}}$ to floating-point precision.

3. **Pucci continuous-yield reconciliation.** For a small coupon and short tenor, set $C_i = c\,P_0\,\Delta_i$ with a continuous coupon yield c. Check that (1) agrees with $P_0\,e^{(r_{\text{repo}}-c)(T_{\text{del}}-T_0)}$ to first order in $r_{\text{repo}}(T_{\text{del}} - T_0)$.

4. **Pucci 24 Jan 2019 anchor.** Reproduce: $P_0 = 103.4922$, 3m repo 2.46%, forward dirty $B_f + A_{\text{del}} = 104.74$, forward clean $B_f = 103.36$. The library's forward must agree to within a fraction of a price tick.

5. `BOND_FROM_QUOTE` **flag.** Verify the library substitutes the market quote into P_0 when the flag is set, and falls back to the bond-curve clean price otherwise. Cross-check that the resulting B_f values differ by exactly the quote–curve gap divided by $D_{\text{repo}}(T_0, T_{\text{del}})$.

8.2 Forward yield

6. **Internal consistency.** Solve (2) for y_f given B_f, then evaluate $B(y_f)$ on the post-delivery cash flows. Check $B(y_f) - B_f \approx A_{\text{del}}$.

7. **Pucci 24 Jan 2019 anchor on y_f.** Reproduce $y_f = 2.530\%$ given the forward dirty 104.74, post-delivery coupons of the cited bond, and a maturity of 25 Nov 2028.

8. **Yield-curve consistency.** If the bond zero-yield curve is bootstrapped from the bond's market price, y_f implied from (2) should be close (but not identical) to the forward zero rate at T_{del} – the difference is the rolldown of the par-yield curve into the forward.

8.3 Clean-price T-Lock

9. **Trade-inception zero NPV.** At $t = T_0$ with $P_{\text{T-Lock}} = B_f$, the NPV from (3) must be exactly zero. This is the standard "fair-strike" check for a forward.

10. **Linearity in price.** A 1bp price move in B_f must produce an NPV change of $-N \times 0.0001 \times D_{\text{T-Lock}}(t, T_{\text{del}})$. Falsify the library if it does not.

11. **Discount curve.** Set $D_{\text{T-Lock}} = 1$ (zero rates). $\text{NPV}_{\text{T-Lock}} = N(P_{\text{T-Lock}} - B_f)$ exactly.

8.4 Yield T-Lock

12. $\text{PV01}(y_f)$ **symmetry.** Equation (4) is a centred difference. Verify it agrees with the analytical modified-duration form $-\partial B/\partial y$ at y_f to first order in the bump size.

13. **Trade-inception zero NPV.** At $t = T_0$ with $y_{\text{T-Lock}} = y_f$, NPV from (5) must be zero.

14. **Linearity in yield.** For $|y_f - y_{\text{T-Lock}}|$ small (a few basis points), the NPV must be linear in $(y_f - y_{\text{T-Lock}})$ with slope $M\,|\text{PV01}(y_f)|\,D_{\text{T-Lock}}(t, T_{\text{del}})$.

15. **Pucci 24 Jan 2019 NPV.** $M = 1$, $y_f = 2.530\%$, $y_{\text{T-Lock}} = 2.717\%$. Compute $|\text{PV01}(y_f)|$ from the bond's post-delivery cash flows, multiply, and discount. Confirm the result matches the order of magnitude in Section 7 above ($\approx -1.59 \times 10^{-4}$ per unit), and that the sign agrees with Pucci's analysis.

16. **Reconciliation with Pucci's exact value.** Run the present paper's formula and Pucci's full forward-contract overhedge on the same trade. The difference is the convexity correction. For a 3m T-Lock at the money this should be of order 10^{-2} per unit notional or smaller; for a 12m at-the-money lock it can be several times larger. Plot the discrepancy vs. tenor and vs. $|y_f - y_{\text{T-Lock}}|$.

8.5 Dirty-price T-Lock

17. **Equivalence to clean-price lock.** The library's dirty-price T-Lock at strike $P^{\text{dirty}}_{\text{T-Lock}}$ must equal its clean-price T-Lock at strike $P^{\text{dirty}}_{\text{T-Lock}} - A_{\text{del}}$ to machine precision.

8.6 Sensitivities and curve bumps

18. **Repo-curve bump.** Bump the repo curve by 1bp parallel. B_f changes; $D_{\text{T-Lock}}$ does not. The NPV change must equal $\partial \text{NPV}_{\text{T-Lock}}/\partial B_f \times \Delta B_f$ from (1).

19. **Bond-yield-curve bump.** Bump the bond zero-yield curve by 1bp parallel. $D_{\text{T-Lock}}$ changes; B_f does not (as it is repo-determined). For the yield-lock, y_f also changes via (2). Verify the library's bumped NPV agrees with a hand-computed total derivative.

20. **Underlier delta limitation.** Compute the library's NDELTA on the underlying bond. Document the gap between this and the true T-Lock delta – the paper itself flags that NDELTA is an underlier-related approximation. The validation report should quantify this gap rather than treating NDELTA as the truth.

21. **Time decay.** Hold all market data fixed and roll the valuation date forward one day. The NPV change should match $-\partial \text{NPV}_{\text{T-Lock}}/\partial t$, which has contributions from B_f rolling toward P, accrued interest changing, and the discount factor $D_{\text{T-Lock}}$ shortening.

8.7 Edge cases and stability

22. **Same-day delivery.** $T_{\text{del}} = T_0$. $B_f = P_0$, $D_{\text{T-Lock}} = 1$. NPV reduces to $N(P_{\text{T-Lock}} - P_0)$ for the clean lock and to $M(y - y_{\text{T-Lock}})|\text{PV01}(y)|$ for the yield lock at the spot yield. Library should handle this without numerical instability.

23. **Coupon date = delivery date.** If $T_N = T_{\text{del}}$, the N-th coupon is paid at delivery. Ensure the library does not double-count C_N: it appears in the sum with $D_{\text{repo}}(T_{\text{del}}, T_{\text{del}}) = 1$, and the bond delivered at T_{del} should be ex-coupon.

24. **Special / on-the-run repo.** For an OTR bond trading special, r_{repo} falls below GC. The forward B_f rises (cost of carry is lower). Confirm the library reproduces the GC-special spread propagation through to NPV: a 100bp specialness on a 3-month forward shifts B_f by approximately $P_0 \times 100\text{bp} \times 0.25 \approx 0.26$ price points.

9 Notation Cheat-Sheet

Symbol	Meaning
P_0	Spot clean price of the underlying bond
A_0, A_{del}	Accrued interest at spot T_0 and at delivery T_{del}
C_i	i-th coupon paid at T_i, $T_0 < T_i \leq T_{\text{del}}$
$D_{\text{repo}}(s, T)$	Repo discount factor from T back to s
$D_{\text{T-Lock}}(t, T)$	T-Lock discount factor from T back to t, on the bond zero-yield curve
B_f	Bond forward (clean) price at delivery, Eq. (1)
y_f	Forward bond yield, Eq. (2)
$P_{\text{T-Lock}}, y_{\text{T-Lock}}$	Locked clean price / locked yield (the strike)
N	Bond quantity (clean-price lock)
M	Trade amount (yield lock)
$B(y)$	Bond present value at yield y
$\text{PV01}(y_f)$	Centred-difference forward-yield risk factor, Eq. (4)

10 Bibliographic Cross-References

- Anonymous (n.d.). *Treasury Lock Model.* Unpublished manuscript. (The paper summarised here.)
- Pucci, A. (2019). *Valuation and Risks of Treasury Locks.* SSRN 3386521. (Companion paper; framework derivation, forward-contract overhedge, convexity analysis.)
- Pucci, A. (2014). *Constant Maturity Treasury Convexity Correction.* SSRN 3387961. (Same family of bond-yield forward IRR machinery, with credit-risk-adjusted discount.)
- Anonymous (n.d.). *Why We Don't Discount at the "Risk-Free Rate".* Unpublished manuscript. (Drift-versus-discount framework underlying the two-curve structure used here.)
- Lou, W. (2018). *TRS pricing under different collateralization schemes.* SSRN 3217420. (Repo as funding for the security leg; CSA-discount choice for $D_{\text{T-Lock}}$.)
- Hedging Future Bond Issuance; Valtchev; "Locking in Treasury Rates with Treasury Lock" – the trade-rationale literature cited in Pucci (2019).